

Prescribed even cycle lengths in cubic graphs

Theorem 1 (Prescribed even lengths). *Let G be a finite simple cubic graph on $n \geq 64$ vertices, with girth g , and put*

$$h := \left\lfloor \frac{g-1}{2} \right\rfloor.$$

For every even integer ℓ satisfying

$$\ell \geq 2 \log_2 n + 4 \quad \text{and} \quad 2^{2h} > \frac{9}{2} n \ell^2, \quad (1)$$

G contains a simple cycle of length exactly ℓ . Equivalently, if $C(G)$ is the set of lengths of simple cycles in G , then

$$\left[2 \log_2 n + 4, \frac{\sqrt{2}}{3} \frac{2^h}{\sqrt{n}} \right) \cap 2\mathbb{Z} \subseteq C(G). \quad (2)$$

The graph need not be connected. All logarithms displayed with subscript 2 are to base 2.

Proof. Fix an even integer ℓ satisfying (1). Write $G = (V, E)$ and $m := |E| = 3n/2$. A cycle is simple, and its length is its number of edges. The definition of h gives

$$1 \leq h < g, \quad g - 2 \leq 2h < g. \quad (3)$$

1. Counting strictly nonbacktracking closed walks. Let $\vec{E}$ be the set of $2m = 3n$ directed edges obtained by giving each edge of G its two orientations. For $e \in \vec{E}$, let $\bar{e}$ be its reverse. Define the nonbacktracking matrix B , with rows and columns indexed by $\vec{E}$, by

$$B_{e,f} := \begin{cases} 1, & \text{if } \text{head}(e) = \text{tail}(f) \text{ and } f \neq \bar{e}, \\ 0, & \text{otherwise.} \end{cases}$$

Set $T_\ell := \text{tr}(B^\ell)$. Expanding the trace gives

$$T_\ell = \sum_{e_0, \dots, e_{\ell-1} \in \vec{E}} \prod_{j=0}^{\ell-1} B_{e_j, e_{j+1}}, \quad e_\ell := e_0. \quad (4)$$

Thus T_ℓ counts ordered directed-edge sequences describing closed walks of length ℓ with no immediate reversal, including between the last and first edges. These will be called *rooted oriented strictly nonbacktracking closed walks*. The first directed edge fixes the root and the direction of traversal; sequences are not identified under rotation or reversal. In vertex notation, such a sequence is $(x_0, \dots, x_\ell = x_0)$ with $x_{j-1} \neq x_{j+1}$ for every j read modulo ℓ .

Let A be the adjacency matrix of G . We will use the identity

$$\det(zI_{2m} - B) = (z^2 - 1)^{m-n} \det(z^2 I_n - zA + 2I_n). \quad (5)$$

This is the cubic case of the Ihara–Bass determinant identity; we derive it here.¹ Define $n \times 2m$ matrices S, H and a $2m \times 2m$ matrix J by

$$S_{v,e} = \mathbf{1}_{\{v=\text{tail}(e)\}}, \quad H_{v,e} = \mathbf{1}_{\{v=\text{head}(e)\}}, \quad J_{e,f} = \mathbf{1}_{\{f=\bar{e}\}},$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator of the stated condition. Directly from these definitions and the fact that every degree is 3,

$$B = H^\top S - J, \quad SH^\top = A, \quad SJH^\top = 3I_n, \quad J^2 = I_{2m}.$$

¹For the general identity and an incidence-matrix proof, see M. Kempton, *Non-backtracking random walks and a weighted Ihara's theorem*, arXiv:1603.05553 (2016), §2.4.

Since J exchanges the two orientations of each edge, for $z \neq \pm 1$ we have

$$\det(zI_{2m} + J) = (z^2 - 1)^m, \quad (zI_{2m} + J)^{-1} = \frac{zI_{2m} - J}{z^2 - 1}.$$

For an invertible square matrix M and compatible rectangular matrices U, V , the determinant formula

$$\det(M - UV) = \det(M) \det(I - VM^{-1}U)$$

follows by eliminating either off-diagonal block in $\begin{pmatrix} M & U \\ V & I \end{pmatrix}$. Applying it with $M = zI_{2m} + J$, $U = H^\top$ and $V = S$ yields

$$\begin{aligned} \det(zI_{2m} - B) &= \det(zI_{2m} + J - H^\top S) \\ &= (z^2 - 1)^m \det(I_n - S(zI_{2m} + J)^{-1} H^\top) \\ &= (z^2 - 1)^m \det\left(I_n - \frac{zA - 3I_n}{z^2 - 1}\right) \\ &= (z^2 - 1)^{m-n} \det(z^2 I_n - zA + 2I_n). \end{aligned}$$

Since $m - n = n/2 \geq 0$, both sides are polynomials in z ; equality for $z \neq \pm 1$ therefore proves (5) for every z . No connectedness assumption has been used.

Let $\lambda_1, \dots, \lambda_n$ be the real eigenvalues of the symmetric matrix A , counted with multiplicity. Because $A\mathbf{1} = 3\mathbf{1}$, take $\lambda_1 = 3$; this remains valid for a disconnected graph. Formula (5) shows that the eigenvalues of B , with algebraic multiplicity, consist of the roots $\mu_{i,+}, \mu_{i,-}$ of

$$\mu^2 - \lambda_i \mu + 2 = 0 \quad (1 \leq i \leq n),$$

together with $m - n$ additional copies of each of 1 and -1 . For any square matrix, triangularization over $\mathbb{C}$ shows that the trace of its ℓ th power is the sum of the ℓ th powers of its eigenvalues, with algebraic multiplicity. In particular, this assertion does not require B to be diagonalizable.

The roots associated with $\lambda_1 = 3$ are 2 and 1. If $|\lambda_i| \geq 2\sqrt{2}$, its two roots are real, and their ℓ th powers are nonnegative because ℓ is even. If $|\lambda_i| < 2\sqrt{2}$, the roots are complex conjugates with modulus $\sqrt{2}$, so

$$\mu_{i,+}^\ell + \mu_{i,-}^\ell = 2 \operatorname{Re}(\mu_{i,+}^\ell) \geq -2 \cdot 2^{\ell/2}.$$

The additional eigenvalues 1 and -1 also contribute nonnegatively. Keeping the contribution 2^ℓ from the first pair and bounding each remaining pair from below gives

$$T_\ell \geq 2^\ell - 2(n-1)2^{\ell/2} > 2^\ell - 2n \cdot 2^{\ell/2}. \quad (6)$$

Since $\ell \geq 2 \log_2 n + 4$, we have $2^{\ell/2} \geq 4n$, whence

$$2n \cdot 2^{\ell/2} \leq 2^{\ell-1}.$$

Consequently,

$$T_\ell > 2^{\ell-1}. \quad (7)$$

2. Bounding internally nonbacktracking returns. For a vertex v and an integer $r \geq 1$, let $f_r(v)$ count the ordered vertex sequences

$$(y_0, y_1, \dots, y_r), \quad y_0 = y_r = v,$$

whose consecutive vertices are adjacent and which satisfy

$$y_{j-1} \neq y_{j+1} \quad (1 \leq j \leq r-1).$$

These are *internally nonbacktracking returns*; no condition is imposed between the last edge and the first edge. We claim that

$$f_r(v) = 0 \quad (1 \leq r < g), \quad f_r(v) \leq \frac{3}{2} 2^{r-h} \quad (r \geq g). \quad (8)$$

Indeed, every internally nonbacktracking walk that repeats a vertex contains a simple cycle. To see this, choose the smallest index j at which a vertex repeats, and let $i < j$ be its earlier occurrence. All vertices preceding index j are distinct. Hence the segment from index i to index j has distinct vertices apart from its identical endpoints. Its length is neither 1, which would be a loop, nor 2, which would be an immediate reversal. It is therefore a simple cycle. It follows that every internally nonbacktracking walk of length less than g is a simple path, and in particular no positive-length return of length less than g exists. This proves the first assertion in (8).

For $r \geq g$, split each return counted by $f_r(v)$ into a prefix of length $r - h$ and a suffix of length h . There are exactly

$$3 \cdot 2^{r-h-1}$$

internally nonbacktracking walks of length $r - h$ starting at v : there are three choices at the first step and two at each later step. Here $r - h \geq 1$ by (3). Fix such a prefix and let its endpoint be w . Every admissible suffix is a simple path from w to v , since $h < g$. If $w = v$, no such positive-length simple path exists. If $w \neq v$, there is at most one such path of length h : the union of two distinct simple paths between the same endpoints contains a cycle and has at most $2h < g$ edges, contradicting the girth. Any nonbacktracking requirement at the prefix-suffix join can only remove possibilities. Each prefix thus has at most one admissible suffix, proving

$$f_r(v) \leq 3 \cdot 2^{r-h-1} = \frac{3}{2} 2^{r-h}.$$

3. Bounding the nonsimple strictly nonbacktracking walks. Let D_ℓ count the walks in T_ℓ for which a vertex repeats among $x_0, \dots, x_{\ell-1}$. Let c_ℓ be the number of simple cycles of length ℓ , identified under cyclic rotation and reversal. Every such cycle has exactly ℓ distinct choices of starting vertex and two choices of direction. Thus

$$T_\ell = 2\ell c_\ell + D_\ell. \quad (9)$$

Consider a walk counted by D_ℓ . Choose indices $0 \leq i < j \leq \ell - 1$ with $x_i = x_j$, and extend its vertex sequence periodically by $x_{k+\ell} = x_k$ for all integers k . Set $v = x_i$ and $r = j - i$. The two segments

$$R_1 = (x_i, x_{i+1}, \dots, x_j), \quad R_2 = (x_j, x_{j+1}, \dots, x_{i+\ell})$$

are returns to v of positive lengths r and $\ell - r$. Both are internally nonbacktracking: every internal consecutive edge pair is a consecutive pair in the original cyclic walk. In particular, the original last-first pair remains nonbacktracking when it lies inside R_2 . By (8),

$$g \leq r \leq \ell - g. \quad (10)$$

The artificial closing pair of either return may reverse, which is permitted by the definition of $f_r(v)$.

For each fixed v and r , there are at most $f_r(v)f_{\ell-r}(v)$ ordered pairs (R_1, R_2) . A pair determines a concatenated closed vertex sequence. Placing its starting point at one of the ℓ edge positions produces at most ℓ rooted oriented sequences, and every walk counted by D_ℓ is produced by some such pair and position. The orientation is already specified by the ordered returns, so there is no additional factor of two. Some concatenations have inadmissible joins, and some walks have multiple representations; retaining all of them can only enlarge this upper bound. For a periodic walk, different starting positions can give the same sequence, which is why the number of resulting sequences is *at most* ℓ . Consequently,

$$\begin{aligned} D_\ell &\leq \ell \sum_{r=g}^{\ell-g} \sum_{v \in V} f_r(v) f_{\ell-r}(v) \\ &\leq \ell \cdot \ell \cdot n \cdot \frac{9}{4} 2^{\ell-2h} = \frac{9}{4} n \ell^2 2^{\ell-2h}. \end{aligned} \quad (11)$$

If $\ell < 2g$, (10) is impossible, so $D_\ell = 0$ and the sum is empty. Otherwise there are at most ℓ possible integer values of r , and (8) bounds each product in the sum by

$$\left(\frac{3}{2} 2^{r-h}\right) \left(\frac{3}{2} 2^{\ell-r-h}\right) = \frac{9}{4} 2^{\ell-2h}.$$

Thus (11) holds in all cases.

4. Extracting a simple cycle. Combining (7), (9) and (11), we obtain

$$\begin{aligned} 2\ell c_\ell &= T_\ell - D_\ell \\ &> 2^{\ell-1} - \frac{9}{4} n\ell^2 2^{\ell-2h} \\ &= 2^{\ell-1} \left(1 - \frac{9n\ell^2}{2^{2h+1}}\right) > 0. \end{aligned}$$

The last inequality is exactly the second hypothesis in (1). Hence $c_\ell > 0$, and G contains a simple cycle of length ℓ . Since ℓ was an arbitrary even integer satisfying (1), this proves the asserted conclusion for every such length. Finally, for $\ell > 0$,

$$2^{2h} > \frac{9}{2} n\ell^2 \iff \ell < \frac{\sqrt{2}}{3} \frac{2^h}{\sqrt{n}},$$

which gives the equivalent interval formulation (2). $\square$

Corollary 2 (The original girth condition). *Every finite simple cubic graph on $n \geq 64$ vertices with girth*

$$g \geq \log_2 n + 2 \log_2(\log_2 n) + 9$$

contains a simple cycle of every even length ℓ satisfying

$$2 \log_2 n + 4 \leq \ell < \frac{16}{3} \log_2 n. \quad (12)$$

In particular, it contains a cycle whose length is a power of two.

Proof. Put $t := \log_2 n \geq 6$ and $h := \lfloor (g-1)/2 \rfloor$. The girth assumption and $2h \geq g-2$ give

$$2^{2h} \geq 2^{g-2} \geq 2^{t+2 \log_2 t+7} = 128nt^2. \quad (13)$$

For every even integer ℓ with $2t+4 \leq \ell < (16/3)t$,

$$\frac{9}{2} n\ell^2 < \frac{9}{2} n \left(\frac{16}{3}t\right)^2 = 128nt^2 \leq 2^{2h}.$$

Both conditions of Theorem 1 therefore hold, proving (12).

To obtain a power-of-two length, let

$$q := 2^{\lceil \log_2(2t+4) \rceil},$$

the least power of two not smaller than $2t+4$. Since $u \leq \lceil u \rceil < u+1$ for every real u ,

$$2t+4 \leq q < 4t+8 \leq \frac{16}{3}t,$$

where the final inequality is equivalent to $t \geq 6$. Also $q \geq 2t+4 \geq 16$, so q is even. Thus q belongs to the interval in (12), and the first part of the corollary supplies a simple cycle of length q . $\square$